

Unit –IV Semiconductors

Semiconductors:

A Semiconductor is a substance whose conductivity or resistivity (10^{-4} to 0.5 ohm -m) lies in between conductors and insulators.

Examples: Si, Ge, GaAs etc.

Properties:

- They have crystalline structure.
- They are formed by covalent bonds.
- They have negative temperature of coefficient of resistance.
- At $T=0\text{K}$, they behaves as insulators.
- When the impurities (III group or V group elements) added to the pure semiconductor, then electrical conductivity increases.

Intrinsic semiconductors and extrinsic semiconductors

Based on purity semiconductors are classified into types,

1. Intrinsic semiconductors
2. Extrinsic semiconductors

Intrinsic semiconductor:

A semiconductor in an extremely pure form is known as intrinsic semiconductor.

Examples: Si, Ge etc.

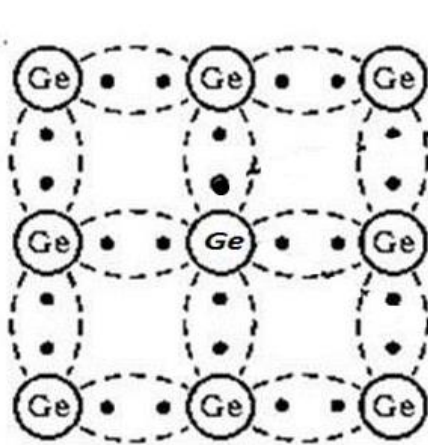
Explanation:

Consider a pure (semi conductor) Si or Ge Crystal. We know that Ge is a forth group element, i.e., it has four valence electrons; to get stability each silicon atom will make four covalent bonds with neighboring Silicon atoms as shown in fig.

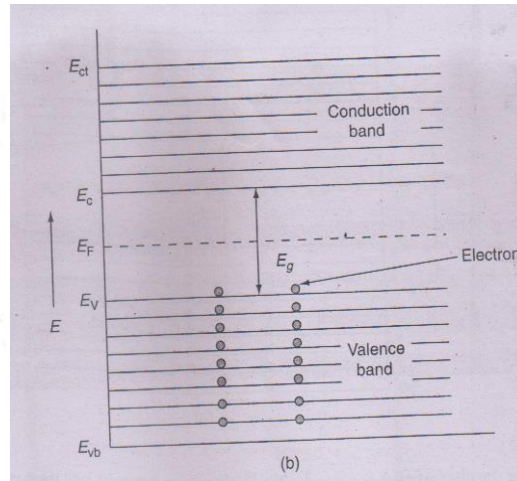
Effect of temperature on intrinsic semiconductor:

i) At absolute zero temperature(0K):

At absolute zero temperature, all the electrons are tightly bound by the semiconductor atoms. The inner orbit electrons are bound to the nucleus where as the valence electrons are engaged in covalent bonding. At this temperature, the covalent bonds are very strong and there are no free electrons. *Therefore, the semi conductor crystal behaves as a perfect insulator at absolute zero temperature.*



a) Structure



b) Energy band diagram

Fig: At absolute zero temperature (0K)

ii) Above absolute zero (> 0 k)

When the temperature is raised, some of the covalent bonds in the semi conductor break due to the thermal energy supplied. The breaking of bonds set those electrons free, at the same time vacant site produced in the covalent bond called “hole”. The result is that a few free electrons exist in the semi conductor. These free electrons can constitute a tiny electric current if potential difference is applied across the semi conductor as shown in fig(c)

Fig (d) shows the energy band diagram. As the temperature is raised, some of the valence electrons acquire sufficient energy to enter into the conduction band and thus become free electrons. Under the influence of electric field, these free electrons will constitute electric current. It may be noted that each time a valence electron enters into the conduction band; a hole is created in the valence band, which acts a positive charge. For one electron set free one hole is created. Therefore, thermal energy creates hole- electron pairs;

In an intrinsic semi conductor, even at room temperature electron-hole pairs are created. Every free electron creates one hole in the valence band. *Therefore the number of electrons in the conduction band and the number of holes in the valence band become equal.*

When electric field is applied across an intrinsic semi conductor, the current conduction takes place by two processes namely: by free electrons and holes. Therefore the total current inside the semiconductor is the sum of currents due to free electrons and holes.

$$I = I_{\text{Electron}} + I_{\text{Hole}}$$

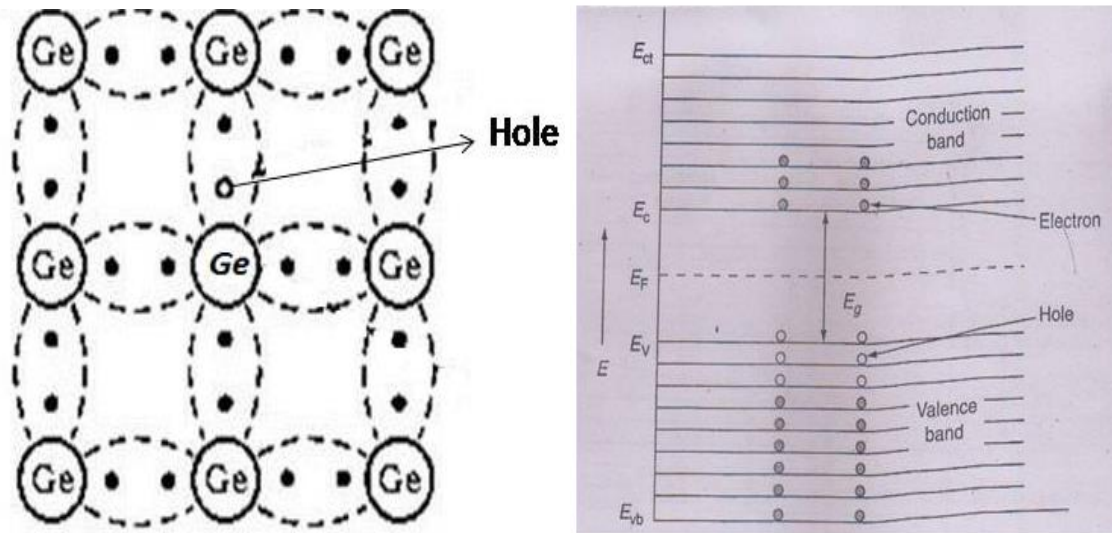


Fig: At $T > 0K$

Extrinsic Semi Conductor:

The electrical conductivity of a pure semiconductor is very small. To increase the conductivity, impurities are added. The impurity added semi conductor is called “**Extrinsic Semi Conductor**”.

(OR)

The conductivity of intrinsic semiconductor can be increased enormously by adding small amounts of impurity atoms [such as III or V group atoms]. This impure semiconductor is called “**Extrinsic Semi Conductor**”.

Doping:

The process of adding impurities to a semiconductor is called doping and the impurity added is called “**Dopant**”.

Depending upon the type of impurity added, Extrinsic Semiconductors are classified into two types.

1. N - Type Semi Conductor.
2. P - Type Semi Conductor.

N – Type Semi Conductor:

When a small quantity of pentavalent impurity is added to a pure semi conductor, it is known as ‘n – type semi conductor’.

The logical relation is

Pure Semi Conductor + V Group Element = n – type Semi Conductor**Explanation:**

- To explain the formation of n-type semi conductor, consider a pure (semi conductor) Germanium Crystal.
- We know that germanium atom has four valence electrons, when a small amount of pentavalent impurities like arsenic (As) is added to a germanium crystal, a large number of free electrons available in the crystal.
- The reason is simple. Arsenic is pentavalent i.e., its atom has five valance electrons. The four of the five valence electrons of Arsenic will make covalent bonds with four germanium atoms. The fifth valence electron of arsenic atom finds no place in covalent bonds and is thus free as shown in fig.
- Therefore for each arsenic atom is added, one free electron will be available in the germanium crystal.
- Thus, a small amount of arsenic impurities (As) are added to a germanium crystal, a large number of free electrons available in the crystal.
- Pentavalent impurities are also known as donor impurities because they donate free electron to semi conductor crystal.

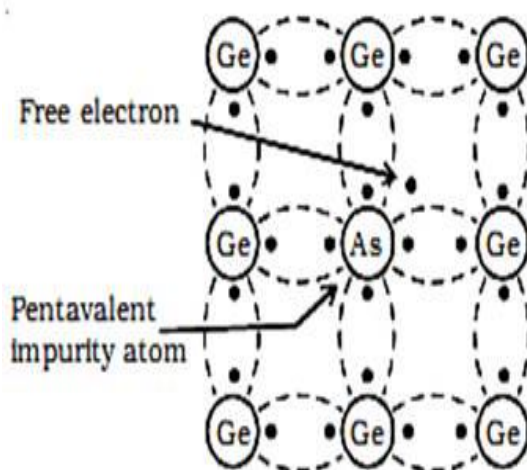
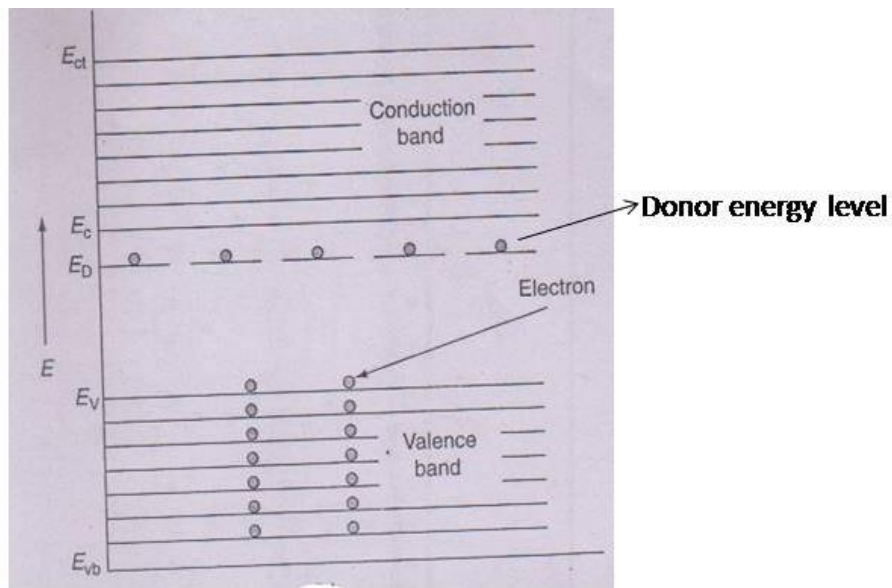
**Energy Band Description:**

Fig (2) shows the energy band description of n-type semi conductor.

The addition of donor impurity to an intrinsic semi conductor creates extra energy level (called donor energy level) just below the bottom of conduction band.

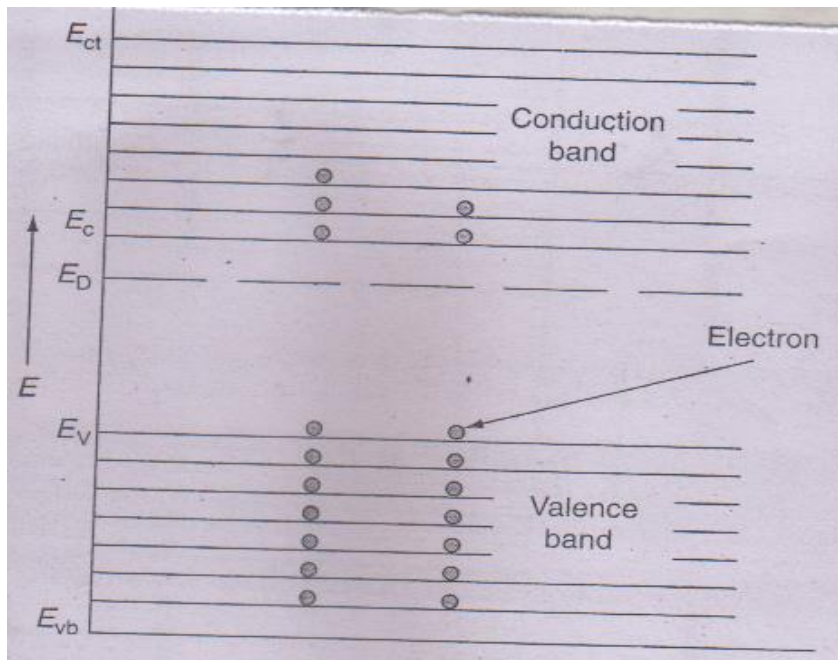
1. At Absolute Zero:

At absolute zero temperature the fifth electrons of arsenic atoms occupy the donor level (represented by dashed lines) as shown in fig.



2. At above Absolute Zero:

Even room temperature provides enough thermal energy to push the electrons sitting at donor energy level to conduction band. (The result is that at room temperature, there are a large number of free electrons in the conduction band as shown in fig) and also few of the covalent bonds in the semi conductor break due to the thermal energy applied. Due to the breaking of covalent bond, there will be a few holes formed in the valance band at this temperature. The result is that at room temperature, there are a large number of free electrons in the conduction band compared to the holes in valence band, i.e., Electrons are majority carriers and holes are minority carriers. So this material is called n-type semi conductor.



P-Type Semi Conductor:

When a small quantity of trivalent impurity (such as indium (In) Boron (B), aluminum (Al) etc. is added to a pure semi conductor, it is called P-type semi conductor.

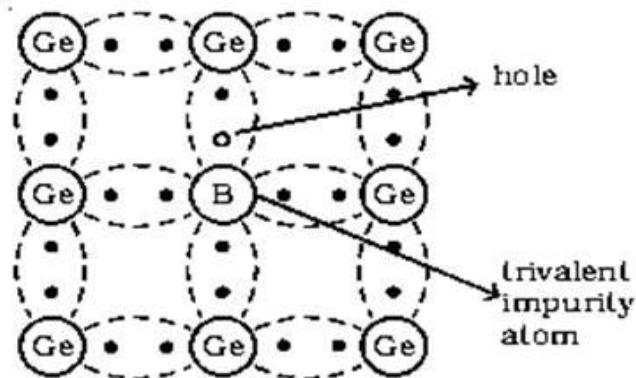
The logical relation is

Pure Semi Conductor + III Group Element = P – type Semi Conductor

Explanation:

- To explain the formation of p-type semi conductor, consider a pure (semi conductor) Germanium Crystal.
- We know that germanium atom has four valence electrons, when a small amount of trivalent impurities like boron atoms (B) are added to a germanium crystal, a large number of holes available in the crystal.
- The reason is simple. Boron is trivalent i.e., its atom has three valence electrons. The three valence electrons of Boron atom will make three covalent bonds with surrounding Ge atoms.
- The fourth bond is incomplete because short of one electron. This missing electron is called a “hole”.
- Therefore for each arsenic atom is added, one free electron will be available in the germanium crystal.
- Thus, a small amount of boron impurities (B) are added to a germanium crystal, a large number of free holes available in the crystal.

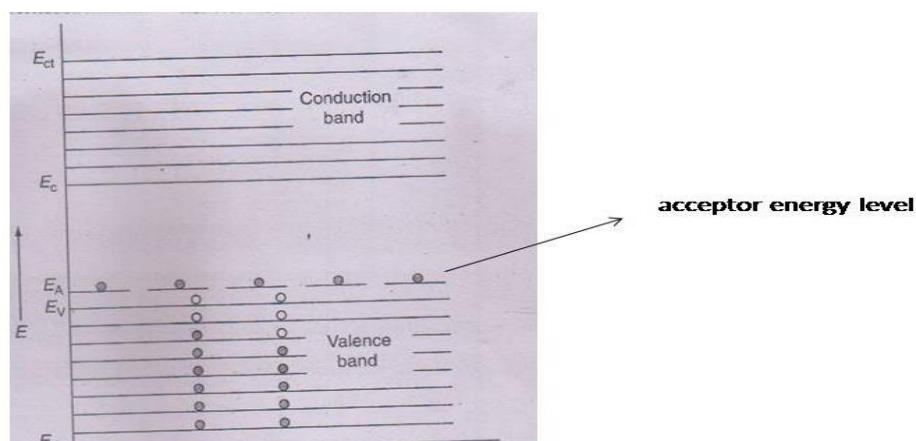
- Trivalent dopant impurities also called acceptors. Because they have need of extra electrons to complete their fourth covalent bonds formation with Ge atoms.



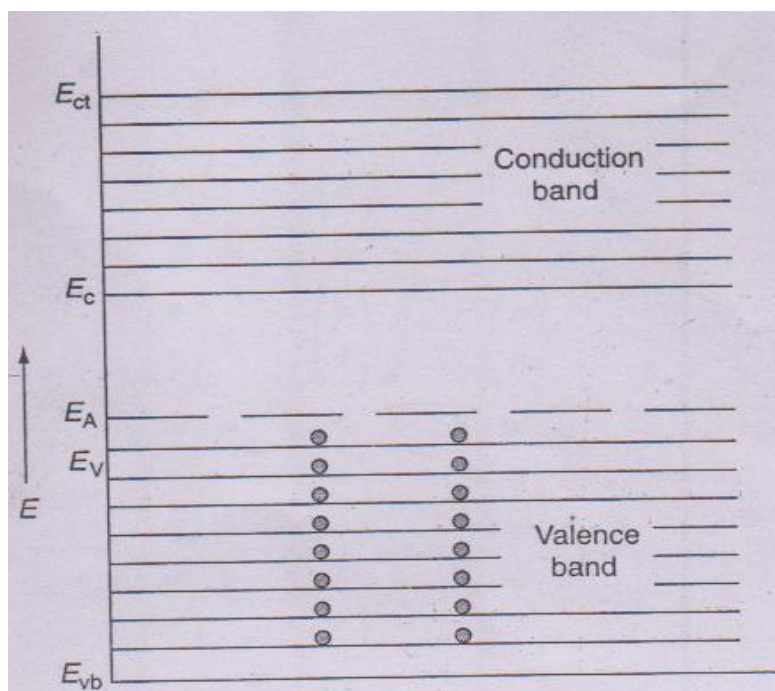
Energy Band Description:

Fig(2) shows the energy band description of p-type semi conductor.

The addition of acceptor impurity to an intrinsic semi conductor creates extra energy level (called acceptor energy level) just above the top of the valance bond [see fig].



Since the energy difference between acceptor level and valence band energy level is very small so even at low temperature the valence band electrons gain sufficient thermal energy to occupy the acceptor energy level states as shown in fig.



After receiving an electron the, Boron atom becomes a negative ion.

But at room temperature provides enough thermal energy to push the electrons in the valence band to the acceptor energy level and also due to thermal agitation a few electrons jumps from V.B to C.B. The result is that at room temperature there are a large number of holes in the V.B compared to the electrons in the C.B. i.e., holes are majority carriers and electrons are minority carries. So, this material is called P-type semi conductors.

Carrier concentration:

In general, the number of charge carriers (free electron or holes) per unit volume of a semi conductor material is called “carrier concentration or density.”

We know that, in the semiconductors, free electron and holes are the charge carriers. So, the carrier concentration can be divided into two types.

1. Electrons carrier concentration and
2. Holes carrier concentration.

Electrons carrier concentration :

The number of free electrons present per unit volume of conduction band of the semi conductor is called “electros carrier concentration or density.”

Let ‘n’ be the electron carrier concentration in the conduction band and is given by

$$n = 2 \left(\frac{2m_e^* \pi K_B T}{h^2} \right)^{3/2} e^{-(E_C - E_F)/K_B T}$$

$$(Or) \quad n = N_c e^{-(E_C - E_F)/K_B T} \longrightarrow (1)$$

Where $N_c = 2 \left(\frac{2m_e^* \pi K_B T}{h^2} \right)^{3/2}$ is called Pseudo constant

and m_e^* is called effective mass of electron.

Holes carrier concentration:

The number of holes present per unit volume of valence band of the semi conductor is called “holes carrier concentration or density.”

Let P be the holes – carrier concentration in the valence band and is given by

$$p = 2 \left(\frac{2m_h^* \pi K_B T}{h^2} \right)^{3/2} e^{-(E_F - E_V)/K_B T}$$

$$(Or) \quad P = N_V e^{-(E_F - E_V)/K_B T} \longrightarrow (2)$$

Where $N_V = 2 \left(\frac{2m_h^* \pi K_B T}{h^2} \right)^{3/2}$ is a Pseudo constant.

and m_h^* is called effective mass of hole.

Carrier concentration in an intrinsic semi conductor:

We know that, at OK in an intrinsic semi conduct, the valence band is completely filled and conduction band is completely empty. The Fermi level is exactly mid way between the valence band and conduction band as shown in fig.

As the temperature of intrinsic semi conductor is increased, electrons from the valence band get thermally excited to the conduction band. These electrons in the conduction band behave like a free particle with an effective mass m_e^*

Similarly, the holes created by these electrons in the valence band also behave like a free particle with an effective mass m_h^* .

Therefore, at any temperature, the number of free electron (electron carrier concentration) in the conduction band is equal to the number of holes (holes carrier concentration) in the valence band.

$$i.e., \quad n = p = n_i$$

where n_i is called intrinsic carrier concentration.

$$\text{Also the product } n.p = n_i^2 \longrightarrow (3)$$

This is known as law of mass action.

Let 'n' be the electron carrier concentration in the conduction band and is given by

$$n = 2 \left(\frac{2m_e^* \pi K_B T}{h^2} \right)^{3/2} e^{-(E_C - E_F)/K_B T}$$

$$(Or) \quad n = N_c e^{-(E_C - E_F)/K_B T} \longrightarrow (1)$$

Where $N_c = 2 \left(\frac{2m_e^* \pi K_B T}{h^2} \right)^{3/2}$ is called Pseudo constant

and m_e^* is called effective mass of electron.

Let P be the holes – carrier concentration in the valence band and is given by

$$p = 2 \left(\frac{2m_h^* \pi K_B T}{h^2} \right)^{3/2} e^{-(E_F - E_V)/K_B T}$$

$$(Or) \quad p = N_v e^{-(E_F - E_V)/K_B T} \longrightarrow (2)$$

Where $N_v = 2 \left(\frac{2m_h^* \pi K_B T}{h^2} \right)^{3/2}$ is a Pseudo constant.

and m_h^* is called effective mass of hole.

Substituting equations (1) and (2) in (3) gives

$$n_i^2 = \left[N_c e^{-(E_C - E_F)/K_B T} \right] \left[N_v e^{-(E_F - E_V)/K_B T} \right]$$

$$= N_C N_V e^{(-E_C + E_F - E_F + E_V)/K_B T}$$

$$= N_C N_V e^{(-E_C + E_V)/K_B T}$$

$$= N_C N_V e^{-(E_C - E_V)/K_B T}$$

But $E_C - E_V = E_g$

$$n_i^2 = N_C N_V \left(e^{-E_g/K_B T} \right)$$

$$n_i = (N_C N_V)^{1/2} \left(e^{-E_g/K_B T} \right) \longrightarrow (4)$$

This equation represents the intrinsic carrier concentration.

Fermi level in an intrinsic semiconductor :-

At temperature T k, the electrons concentration (n) is equal to the holes concentration

‘P’ in the intrinsic semiconductor

$$\text{i.e. } n = P$$

Therefore [from equations (1) and (2)]

$$N_c e^{-\frac{(E_C - E_F)}{K_B T}} = N_v e^{-\frac{(E_F - E_V)}{K_B T}}$$

$$\frac{e^{-\frac{(E_C - E_F)}{K_B T}}}{e^{-\frac{(E_F - E_V)}{K_B T}}} = \frac{N_v}{N_c}$$

$$e^{-\frac{(E_C - E_F)}{K_B T}} \times e^{\frac{(E_F - E_V)}{K_B T}} = \frac{N_v}{N_c}$$

$$e^{\left(\frac{-E_C + E_F + E_F - E_V}{K_B T}\right)} = \frac{N_v}{N_c}$$

$$e^{\left(\frac{2E_F - (E_C + E_V)}{K_B T}\right)} = \frac{N_v}{N_c}$$

Taking log on both sides, we have

$$\frac{2E_F - (E_C + E_V)}{K_B T} = \log \left(\frac{N_v}{N_c} \right)$$

$$E_F - \frac{(E_C + E_V)}{2} = \frac{1}{2} \log \left(\frac{N_v}{N_c} \right) \cdot K_B T$$

$$E_F - \frac{(E_C + E_V)}{2} = \frac{1}{2} K_B T \log \left(\frac{N_v}{N_c} \right)$$

$$E_F = \frac{E_C + E_V}{2} + \frac{1}{2} K_B T \log \left(\frac{N_v}{N_c} \right) \quad (5)$$

$$\text{If } N_v = N_c \text{ then } \log \left(\frac{N_v}{N_c} \right) = 0$$

Eq (5) becomes

$$E_F = \frac{E_C + E_V}{2}$$

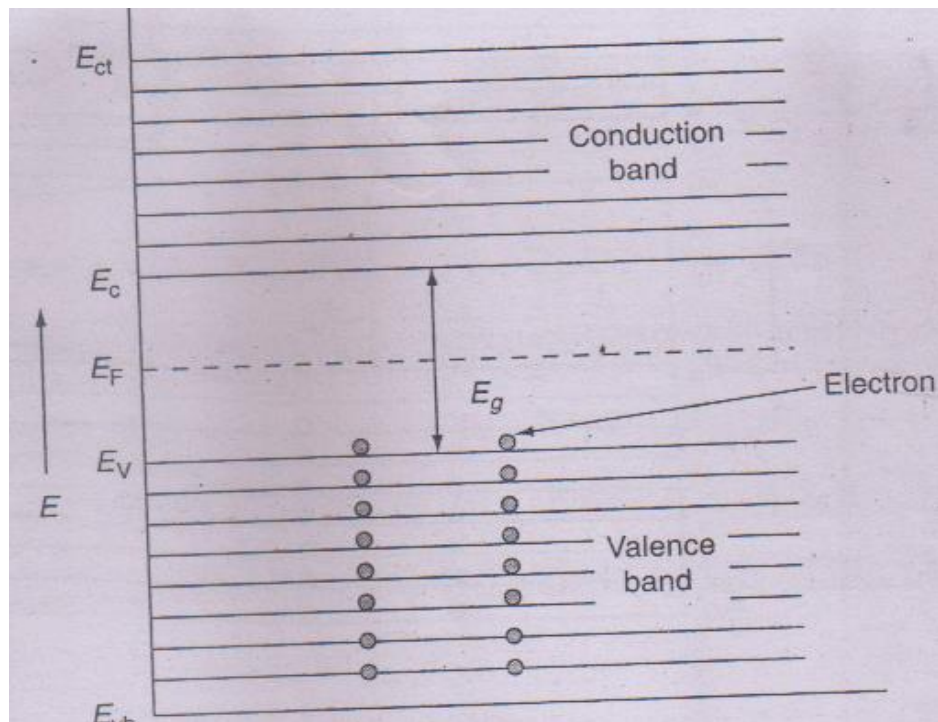
Thus, the Fermi level in an intrinsic semiconductor lies in midway between E_C and E_V .

Dependence of E_F on temperature:

The variation of Fermi level with temperature for intrinsic semiconductor is shown in fig.

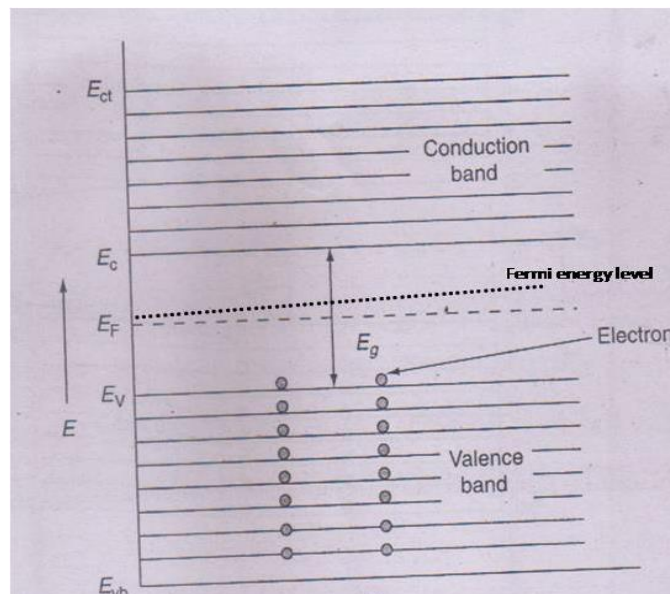
Case (a):

At 0 K, the Fermi level lies exactly in the middle of forbidden gap as shown in fig.



Case(b) :

The Fermi level slightly increases with the increase in temperature as shown in figure.



Electrical conduction in semiconductors

In the semiconductors electrical conduction arises due to two process namely

1. Drift

2. diffusion

Drift:

Definition:-

Under an applied electric field both the charge carriers i.e., electrons and holes acquire an average velocity called the drift velocity (V_d) and this process is called drift.

The drift velocity V_d is small. It is directly proportional to the electrical field E .

$$\text{i.e., } V_d \propto E$$

$$V_d = \mu E \longrightarrow (1)$$

Where μ is a constant and is called the mobility of charge carriers.

If μ_n and μ_p are the electron mobility and hole mobility respectively, then

$$V_n = \mu_n E \text{ and } \longrightarrow (2)$$

$$V_p = \mu_p E \longrightarrow (3)$$

1. We know that, Current $I = neAV_d$.

But in a semi conductor, the total current is the sum of electrons current and holes current.

Let n be the electron concentration and p be the hole concentration.

Therefore $I = I_n + I_p$

$$= neAV_n + pAV_p$$

$$= eA (nV_n + pV_p) \quad [\text{From equations (2) \& (3)}]$$

$$= eA(n\mu_n E + p\mu_p E)$$

$$= eA(n\mu_n + p\mu_p)E$$

2. Current density $J = \frac{I}{A}$

$$J_n = \frac{neA\mu_n E}{A} \text{ and}$$

$$J_p = \frac{peA\mu_p E}{A}$$

Therefore, total current density $J = J_n + J_p$

$$= \frac{eA(n\mu_n + p\mu_p)E}{A} = e(n\mu_n + p\mu_p)E$$

$$\text{Conductivity } \sigma = \frac{J}{E} = \frac{e(n\mu_n + p\mu_p)E}{E} = e(n\mu_n + p\mu_p)$$

NOTE:-

1. For an intrinsic semiconductor: $n=p=n_i$

Where n_i is the carrier concentration of the intrinsic semiconductor.

$$I_{inc} = eA(n_i\mu_n + n_i\mu_p)E$$

$$= eAn_i(\mu_n + \mu_p)E$$

$$\therefore \text{Current density } J_{in} = \frac{I_{in}}{A} = \frac{eAn_i(\mu_n + \mu_p)E}{A} = en_i(\mu_n + \mu_p)E$$

$$\text{Conductivity } \sigma_{in} = \frac{J_{in}}{E} = \frac{en_i(\mu_n + \mu_p)E}{E} = en_i(\mu_n + \mu_p)$$

FOR AN EXTRINSIC SEMICONDUCTOR:-

a) **For n-type semiconductor :-** The hole current is neglected, as the holes are minority carriers and the electrons are the majority carriers.

$n \approx N_d$ is the donor electron concentration

[$\therefore$ Where N_a can be neglected]

$$I = eA V_n N_d$$

$$= eA\mu_n EN_d$$

$$J = \frac{I}{A} = e\mu_n EN_d$$

$$\sigma = \frac{J}{E} = e\mu_n EN_d$$

b) **For P-type semiconductor :-** The electron current is neglected and $P=N_{ai}$, where N_a is the acceptor hole concentration. [$\therefore$ When N_d can be neglected].

$$I = eA V_p N_a$$

$$= eA\mu_p E N_a$$

$$J = \frac{I}{A} = e\mu_p E N_a$$

$$\sigma = \frac{J}{E} = e\mu_p N_a$$

DIFFUSION :-

Diffusion takes place due to existence of non-uniform concentration of charge carriers.

Definition:-

Due to non –uniform carrier concentration in a semiconductor ,the charge carriers moves from a region of higher concentration to a region of higher concentration to a region of lower concentration. This process is known as diffusion of charge carriers.

→ According to Fick's law, the diffusion flux F is directly proportional to concentration gradient [i.e. carrier concentration per unit length] of the particles and is in the opposite direction .i.e;

$$F \propto \frac{-\partial(\Delta N)}{\partial X}$$

$$F = -D \frac{\partial(\Delta N)}{\partial X} \longrightarrow (1)$$

Where D is diffusion constant.

And ΔN is the excess carrier concentration.

Let Δn and Δp be the excess concentration in a semiconductor material.

Then Fick's law can be written as,

$$F \propto \frac{-\partial(\Delta N)}{\partial X} \quad (\text{For electrons})$$

$$F = -D_n \frac{\partial(\Delta N)}{\partial X} \longrightarrow (2)$$

and

$$F \propto \frac{-\partial(\Delta P)}{\partial X} \quad (\text{For electrons})$$

$$F = -D_p \frac{\partial(\Delta P)}{\partial X} \longrightarrow (3)$$

→ But the current density due to diffusion depends on charge of particle and rate of diffusion.

$$\therefore \text{Current density (J)} = \text{Charge of particle} \times \text{Rate of diffusion} \longrightarrow (4)$$

→ The diffusion flux density due to electron is given by

$$J_{n(\text{Diff})} = -e \left[-D_n \frac{\partial(\Delta N)}{\partial X} \right] \quad \therefore \text{from(4)}$$

$$J_{n(\text{Diff})} = e D_n \frac{\partial(\Delta N)}{\partial X} \longrightarrow (5)$$

The diffusion flux density due to hole is given by

$$J_{p(\text{Diff})} = + e \left[D_p \frac{\partial(\Delta P)}{\partial X} \right] \longrightarrow (6)$$

Where +e and –e are charge of the hole and electron respectively.

$\therefore$ The total current density due to diffusion is given by

$$J_{\text{TOTAL, (DIFF)}} = J_{n(\text{diff})} + J_{p(\text{diff})}$$

$$J_{\text{TOTAL, (DIFF)}} = e D_n \frac{\partial(\Delta N)}{\partial X} - e D_p \frac{\partial(\Delta P)}{\partial X} \longrightarrow (7)$$

NOTE:- The total current density due to excess electrons is the sum of the current densities due to drift and diffusion of electrons.

$$J_n = J_{n(\text{drift})} + J_{n(\text{diff})} \\ = \Delta n \mu_n E + e D_n \frac{\partial(\Delta n)}{\partial x} \longrightarrow (8)$$

Similarly,

The total current density due to excess holes is the sum of the holes densities due to drift and diffusion of holes.

$$J_p = \Delta p e \mu_p E - e D_p \frac{\partial(\Delta p)}{\partial x} \longrightarrow (9)$$

EINSTEIN'S RELATION:-

Einstein showed the direct relation between the mobility (μ) and diffusion coefficient (D) of a semiconductor. This relation is called Einstein relation.

At any disturbance in equilibrium, the drift and diffusion current densities due to carrier concentration are equal.

$$\text{i.e.; } J_{n(\text{drift})} = J_{n(\text{diff})} \\ (\Delta n) e E \mu_n = D_n e \frac{\partial(\Delta n)}{\partial x} \longrightarrow (1)$$

The force on excess electrons (Δn) to restore equilibrium is given by the product of Δn and electric forces eE on each electron. Thus,

$$F = (\Delta n) e E \longrightarrow (2)$$

Equation (1) can be written as

$$F \cdot \mu_n = D_n e \frac{\partial(\Delta n)}{\partial x} \\ F = \frac{D_n}{\mu_n} e \frac{\partial(\Delta n)}{\partial x} \longrightarrow (3)$$

From the kinetic theory of gases, the force on gas molecules (charge carriers) is given by

$$F = K_B T \frac{\partial(\Delta n)}{\partial x} \longrightarrow (4)$$

Comparing (3) and (4), we get

$$K_B T \frac{\partial(\Delta n)}{\partial x} = \frac{D_n}{\mu_n} e \frac{\partial(\Delta n)}{\partial x} \\ K_B T = \frac{D_n}{\mu_n} e \\ D_n = \frac{\mu_n k_B T}{e}$$

$$\frac{D_n}{\mu_n} = \frac{k_B T}{e} \longrightarrow (5)$$

Similarly for holes,

$$\frac{D_p}{\mu_p} = \frac{k_B T}{e} \longrightarrow (6)$$

From equations (5) and (6)

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p}$$

$$\frac{D_n}{D_p} = \frac{\mu_n}{\mu_p}$$

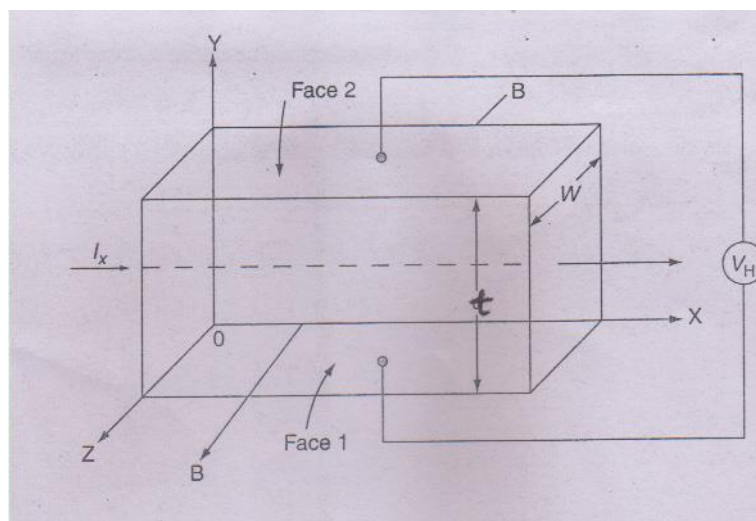
The above relation is known as Einstein's relation.

Hall Effect:

When a current carrying semiconductor is placed in the magnetic field, then a potential difference (voltage) or electric field is developed in the material perpendicular to both the current and the applied magnetic field. This effect is known as Hall Effect and the developed potential difference (voltage) is known as hall voltage.

Explanation:

Consider a semiconductor slab of thickness 't' and width 'w' in which current (I) flowing along X-direction and magnetic field is applied along Z- direction as shown in fig.



The charge carriers inside the semiconductor experience a force due to the applied magnetic field called Lorentz force.

As a result, electrons move (drift) towards downward direction with velocity ' V_d ' and accumulate at bottom surface (face 1) and the corresponding positive charge on upper surface (face2) of the material as shown in fig.

Thus a potential difference and electric field developed across the surface. This developed electric field along Y-direction exerts a force ($F_e = eE_H$) on the electrons to prevent further accumulation at the bottom surface.

Therefore, the force due to potential difference $F_e = eE_H$

the force due to magnetic field $F_L = BeV_d$

At steady state

$$F_e = F_L$$

$$eE_H = Bev_d$$

$$E_H = Bv_d \longrightarrow (1)$$

For n-type:

If ' n ' is the electrons carrier concentration in the semiconductor, then current density

$$J = -neV_d$$

$$V_d = \frac{J}{-ne} \longrightarrow (2)$$

Substituting eq (2) in eq(1)

$$E_H = \frac{BJ}{-ne}$$

$$\boxed{E_H = R_H BJ} \longrightarrow (3)$$

Where $R_H = \frac{1}{-ne}$ is known as Hall coefficient.

For p-type:

If ' p ' is the electrons carrier concentration in the semiconductor, then current density

$$J = peV_d$$

$$V_d = \frac{J}{pe} \longrightarrow (2)$$

Substituting eq (2) in eq(1)

$$E_H = \frac{BJ}{pe}$$

$$\boxed{E_H = R_H BJ} \longrightarrow (3)$$

Where $R_H = \frac{1}{pe}$ is known as Hall coefficient

Hall coefficient in terms of hall voltage:

If the thickness of the sample is 't' and the developed voltage is V_H , then the Hall voltage

$$V_H = E_H \cdot t \longrightarrow (4)$$

Substituting eq (3) in eq(4);

$$V_H = R_H BJ t \longrightarrow (5)$$

If 'w' is the width of the semiconductor,

Then, area of the sample $A = w \cdot t$

But current density $J = \frac{I}{A} = \frac{I}{w t} \longrightarrow (6)$

Substituting eq (8) in eq(7); we get

$$V_H = \frac{R_H IB}{w}$$

$$R_H = \frac{V_H w}{IB}$$

$$\boxed{R_H = \frac{V_H w}{IB}}$$

This is the relation between Hall coefficient and hall voltage.

- The electrical conductivity in a semiconductor is given by

$$\sigma = ne\mu$$

$$\mu = \frac{\sigma}{ne}$$

$$\boxed{\mu = R_H \sigma}$$

APPLICATIONS:

- Knowing R_H and σ , the mobility of charge carriers can be measured.
- It is used to find the carrier concentration

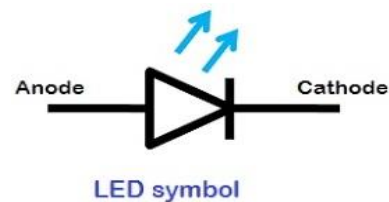
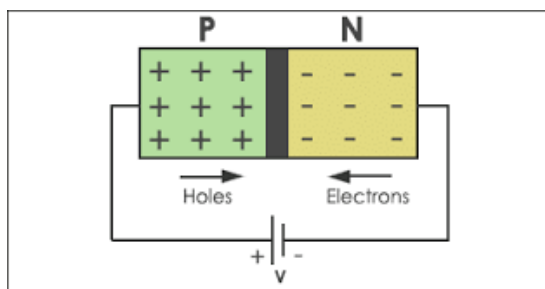
$$\boxed{n = -\frac{1}{eR_H} \quad \text{and} \quad p = \frac{1}{eR_H}}$$

- It is used to determine whether the material is p-type or n-type semiconductor.
i.e., if R_H is negative then the material is n-type.
if R_H is positive then the material is p-type.

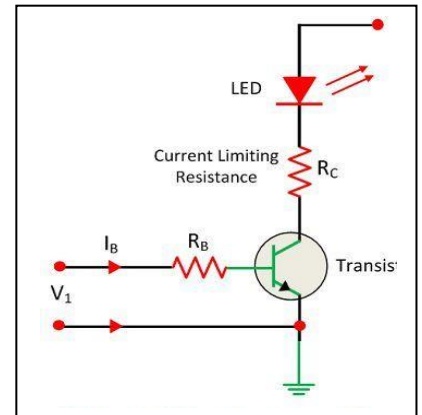
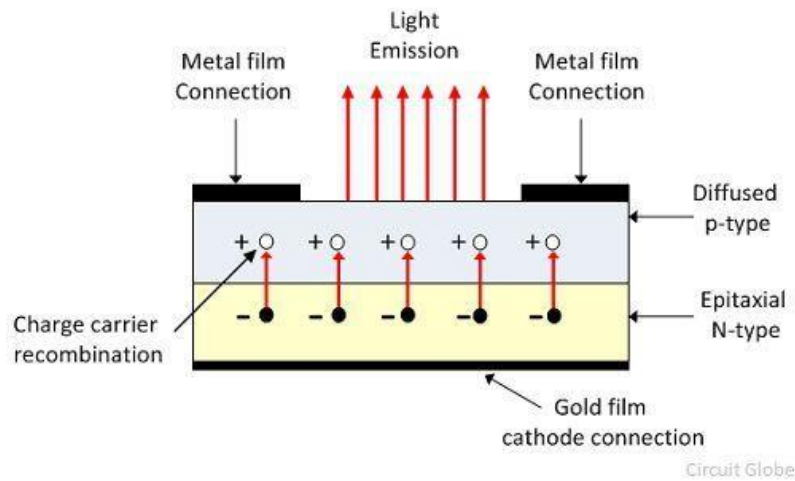
LIGHT EMITTING DIODE (LED):-

Light emitting diode is a forward biased p-n junction device, used to convert the electrical energy into optical energy.

- The LED is a [PN-junction](#) diode which emits light when an [electric current](#) passes through it in the forward direction.
- In the LED, the recombination of charge carrier takes place.
- The electron from the N-side and the hole from the P-side are combined and gives the energy in the form of heat and light.
- The LED is made of [semiconductor](#) material which is colourless, and the light is radiated through the junction of the diode.
- The LEDs are extensively used in segmental and dot matrix displays of numeric and alphanumeric character.
- The several LEDs are used for making the single line segment while for making the decimal point single LED is used.

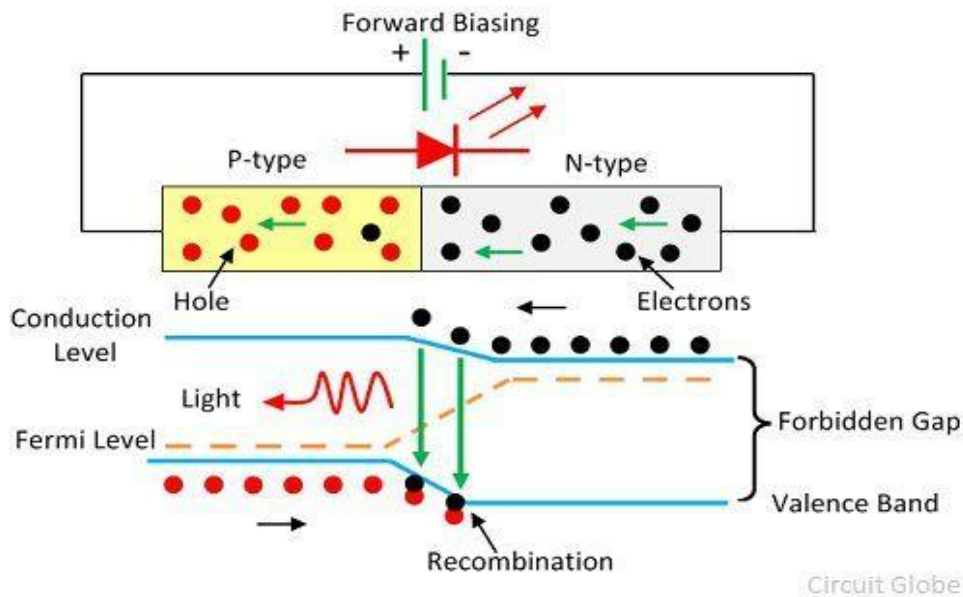


Construction of LED



- The recombination of the charge carrier occurs in the P-type material, and hence P-material is kept at the upper surface of the LED.
- For the maximum emission of light, the anode is deposited at the edge of the P-type material.
- The cathode is made of gold film, and it is usually placed at the bottom of the N-region.
- This gold layer of cathode helps in reflecting the light to the surface.
- The gallium arsenide phosphide is used for the manufacturing of LED which emits red or yellow light for emission. The LED are also available in green, yellow amber and red in colour.
- The simple [transistor](#) can be used for off/on of a LED as shown in the figure .
- The base current I_B conducts the transistor, and the transistor conducts heavily.
- The resistance R_C limits the current of the LED.

Working:-



- The working of the LED depends on the quantum theory.
- The quantum theory states that when the energy of electrons decreases from the higher level to lower level, it emits energy in the form of photons.
- The energy of the photons is equal to the gap between the higher and lower level.
- The LED is connected in the forward biased, which allows the current to flows in the forward direction.
- The flow of current is because of the movement of electrons in the opposite direction.
- The recombination shows that the electrons move from the conduction band to valence band and they emits electromagnetic energy in the form of photons.
- The energy of photons is equal to the gap between the valence and the conduction band.
- Efficiency of generation of light increases with the injected current and with a decrease in temperature
- The light is concentrated near the junction as the carriers are available within a diffusion length of the junction
- LED's radiate different colours such as red, green, yellow, Orange and white
- Some of the LEDs emit Infra red (inVisible)
- Wavelength of emitted light depends upon the energy gap of the material.

Material

Colour

GaAs	Infrared radiation (invisible)
Gap	Green
GaAs p	Red
Al In Gap	Yellow
SiC	Blue
GaN	Blue

Advantages of LED in electronic displays:

- ✓ The followings are the major advantages of the LED in an electronics displays.
- ✓ The LED are smaller in sizes, and they can be stacked together to form numeric and alphanumeric display in the high-density matrix.
- ✓ The intensity of the light output of the LED depends on the current flows through it. The intensity of their light can be controlled smoothly.
- ✓ The LED are available which emits light in the different colours like red, yellow, green and amber.
- ✓ The on and off time or switching time of the LED is less than of 1 nanoseconds. Because of this, the LED are used for the dynamic operation.
- ✓ The LEDs are very economical and giving the high degree of reliability because they are manufactured with the same technology as that of the transistor.
- ✓ The LED are operated over a wide range of temperature say $0^{\circ} - 70^{\circ}$. Also, it is very durable and can withstand shock and variation.
- ✓ The LED have a high efficiency, but they require moderate power for operation. Typically, the voltage of 1.2V and the current of 20mA is required for full brightness. Therefore, it is used in a place where less power are available.

Disadvantages of LED

The LED consumes more power as compared to LCD, and their cost is high. Also, it is not used for making the large display.

Applications

- In motor vehicles and bicycle lights.
- In traffic light Indicators, signs and signals.
- In data displaying boards.

- In medical applications and toys
- Non visual applications.
- In light bulbs and many more.
- Remote controls